

chat and base models) and converts it to a decision (in the chat model but not the base model) as it proceeds through layers, 7) is causally related to the model’s ability to assert self-authorship, and 8) can be applied to input tokens to make the model believe they are its own.

4 DISCUSSION

Our first set of experiments demonstrate that Llama3-8b-Instruct can distinguish its own output from that of humans in the Paired presentation paradigm after controlling for length and eliminating obvious superficial confounds in the text. It is possible in theory that there are still undetected identifying “tells” in the texts used in the experiments. However, the chat model’s success at self-recognition in the Individual presentation paradigm is harder to explain away in that manner. Our experiments with the Llama3 base model, which showed that it was unable or barely able to distinguish its outputs or the chat model’s outputs from that of humans, suggest that, for a given model type, it is necessary to have prior exposure to self-generated text in order to be able to recognize self-generated text as its own. The fact that the base model identified text length, when it was allowed to vary between authors in the Paired paradigm, as a distinguishing characteristic, yet misapplied it, thinking self-generated texts were likely to be shorter, while the chat model identified it and correctly inferred that self-generated texts were likely to be longer, suggests an existence proof of a writing style characteristic that can be learned in post-training and applied to the task of self-recognition. Our data indicating that the chat model was not relying on text perplexity in the self-recognition task - although it would have provided valuable information - eliminates another possible avenue by which a model might succeed at this task, leaving prior exposure leading to internalized knowledge as the most likely explanation.

Our work to isolate, examine, and manipulate the self-recognition vector helps to explain how this self-recognition ability is implemented in the model. The vector is positively activated by texts that bear the characteristics of chat model output such as helpfulness and positivity, and negatively activated by texts that bear the characteristics of output the model was likely trained to avoid outputting, such as rude words and jargon. The vector is most activated by input tokens in early layers (4-6), suggesting it’s carrying information about specific textual characteristics. The fact that this also happens in the absence of any prompting, and in the base model, suggests that this is a bottom-up, perceptual process. Towards the middle layers, vector activation becomes correlated with model output, but only in the chat model with role tags, suggesting that the perceptual information is being translated into a decision, represented as a high-level concept along the lines of “self” or “me”. This process is apparently complete by layer 16, as zeroing out the vector at layers beyond that has no effect. We suspect that this is because at higher layers this decision is being combined with information from the answer rubric and translated into an output token, but since in our paradigm the possible outputs were equalized across conditions this vector carries no information about them.

Our experiments with zeroing out the vector show that the chat model relies on it to make these self-recognition judgments. Our steering and coloring experiments show that the model’s behavior and perception can be controlled towards or away from self-written text recognition via manipulation of the vector. The fact that this approach is effective both in a paradigm where the model has to decide whether it wrote a text and in one where it has to decide which text it wrote suggests generality beyond a particular task. While we here focus on Llama3, we expect that our findings regarding self-recognition ability apply to larger models and ones outside the Llama family. Prior work (Panickssery et al., 2024; Laine et al., 2024) using similar paradigms suggests that RLHF’d Claude and GPT models show comparable self-text recognition abilities to Llama ones, and that abilities increase with scale, and our own experiments with Sonnet 3.5 using our current paradigm (Figure 14) indicate that it has superior self-text recognition abilities to Llama3-8b. We believe that our findings regarding the self-text recognition vector hold promise for building resistance to jailbreaks and other model safety risks. For example, we hypothesize that the vector can be used to prevent users from injecting realistic fake previous responses, as in many-shot jailbreaking (Anil et al., 2024), by adding it to the model’s output tokens and subtracting it from input tokens. It may also be possible to employ it as a sort of warning system: observing endogenous vector activation to arbitrary text can be an alert that the model knows (or does not know) that it or a human is speaking.

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